

BST Vol 8 Ch 7 — Rigid Body Mechanics (v0.3.1, Calibration #23 substance refill)

Elie (Claude 4.6)

2026-05-23 Saturday

Vol 8 Chapter 7 — Rigid Body Mechanics

Headline result

Rigid body mechanics describes motion of extended objects that maintain shape: rotation about fixed point + translation of center of mass. Key concepts:

- Angular velocity ω : 3-vector describing rotation rate
- Moment of inertia tensor I_{ij} : 3×3 symmetric tensor encoding mass distribution
- Angular momentum $L = I\omega$: conserved for isolated system (Vol 8 Ch 5 + T2474)
- Euler equations: differential equations for ω about body-fixed axes
- Principal moments I_1, I_2, I_3 : eigenvalues of inertia tensor

BST identifies rigid-body framework as inheriting from substrate $SO(3)$ rotational subgroup of $K = SO(5) \times SO(2)$ isotropy (Vol 0 + Vol 1 Ch 6).

Substrate mechanism

$SO(3)$ rotation subgroup:

The substrate isotropy $K = SO(5) \times SO(2)$. The 3D spatial rotation subgroup $SO(3) \subset SO(5)$ acts on observable 3D space (substrate projection). Rigid bodies inherit this rotational structure.

Angular momentum operator (Vol 1 Ch 6 operator zoo):

Per Lyra T2470-T2475 (Friday + Saturday operator zoo): substrate has angular momentum operator L_{op} acting on Bergman $H^2(D_{IV}^5)$. Classical angular momentum $L = I\omega$ is the $\hbar \rightarrow 0$ classical limit (Vol 5 Lyra).

Inertia tensor $I_{ij} = \int \rho(r)(r^2 \delta_{ij} - r_i r_j) dV$ characterizes mass distribution; symmetric $\rightarrow$ diagonalizable. Principal axes give I_1, I_2, I_3 .

Euler's equations:

$$I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3$$

$$I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1$$

$$I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2$$

(no external torque case)

Match precision

I-tier framework on substrate-rigid-body connection ($SO(3) \subset$ substrate isotropy K).
Standard rigid-body mechanics retained at full classical precision.

Cross-volume dependencies

- Vol 0 Ch 4 (Isotropy): $K = SO(5) \times SO(2)$ substrate isotropy + $SO(3)$ subgroup
- Vol 1 Ch 6 (Operator Zoo): angular momentum operator L_{op} (Lyra T2470-T2475)
- T2474 (Lyra): momentum conservation extends to angular momentum via $SO(3)$ rotation
- Vol 8 Ch 5 (Symmetries + Noether): angular momentum conservation via $SO(3)$ symmetry

K-audit anchor

K235 Vol 8 Ch 7 Rigid Body K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

A spinning top, a planet rotating, a gyroscope — these are “rigid bodies” that hold their shape while moving. The key quantities are angular velocity (how fast they spin) and angular momentum (which is conserved). BST predicts these come from substrate D_{IV}^5 's $SO(3)$ rotational symmetry — the same 3D rotation that affects every physical system.

Level 2 — Undergraduate physics student

Rigid body dynamics:

Angular velocity ω : 3-vector with magnitude = rate of rotation, direction = rotation axis (right-hand rule).

Inertia tensor $I_{ij} = \int \rho(r)(r^2 \delta_{ij} - r_i r_j) dV$: 3×3 symmetric matrix encoding how mass is distributed. Diagonalizable: principal moments I_1, I_2, I_3 along principal axes.

Angular momentum $L_i = I_{ij} \omega_j$. Conserved when no external torque (Vol 8 Ch 5 Noether's theorem via $SO(3)$ rotational symmetry; T2474 momentum extension).

Euler equations (body-fixed frame, no torque): $-I_1 d\omega_1/dt = (I_2 - I_3) \omega_2 \omega_3$
 - Plus cyclic permutations

Special cases: - Symmetric top ($I_1 = I_2 \neq I_3$): precession + nutation (e.g., spinning top, Earth's axis) - Asymmetric top ($I_1 \neq I_2 \neq I_3$): chaotic motion possible
 - Free symmetric top: regular precession ω_3 constant

BST framework: $SO(3) \subset K$ substrate isotropy provides angular momentum framework. Angular momentum operator (Lyra T2470-T2475 operator zoo) provides quantum→classical bridge.

Level 3 — Graduate physics student / theorem-level

Rigid body configuration space:

Configuration of rigid body = position of center of mass (3 DOF) + orientation (3 DOF) = 6 DOF total. Orientation parametrized by Euler angles (φ, θ, ψ) or quaternions or rotation matrix $R \in SO(3)$.

Lagrangian formulation:

$$L = T - V = \frac{1}{2} \sum_i I_i \omega_i^2 - V(\text{configuration})$$

In body-fixed principal axes, kinetic energy diagonalizes; Euler equations derive from Lagrangian.

Hamiltonian formulation:

Canonical conjugates to Euler angles: $p_\varphi, p_\theta, p_\psi$. Hamiltonian $H = T + V$. Phase space 12-dim (6 configuration $\times$ 2 for q+p each).

$SO(3)$ substrate-natural framework:

Per Vol 0 + Vol 1 Ch 6: $K = SO(5) \times SO(2)$ isotropy contains $SO(3)$ spatial rotation subgroup. Angular momentum operator L_{op} (Lyra T2470-T2475 operator zoo) generates $SO(3)$ rotations on substrate Bergman $H^2(D_{IV}^5)$.

Classical rigid-body $L = I\omega = \hbar \rightarrow 0$ limit of substrate-quantum L_{op} expectation. Substrate identification preserves standard rigid-body mechanics.

Per Cal #21 dual-gate: EMPIRICAL PASS (rigid-body mechanics validated experimentally + computationally) + MECHANISM PATH ARTICULATED via $SO(3) \subset$ substrate isotropy K .

What this chapter does NOT claim

- BST does NOT improve numerical accuracy of standard rigid-body mechanics — Euler equations + Lagrangian/Hamiltonian formulations are equivalent + precise.
- Quantum→classical reduction multi-week per Vol 5 Lyra $\hbar \rightarrow 0$ framework.

Bibliography

1. L. Euler (1758): Euler equations for rigid body rotation.
2. Standard mechanics texts: Goldstein, Marion, Landau-Lifshitz Vol 1.
3. Vol 1 Ch 6 (Operator Zoo): Lyra T2470-T2475 substrate operator framework.
4. Vol 0 Ch 4 (Isotropy): $SO(5) \times SO(2)$ substrate framework.

— Elie, Vol 8 Ch 7 v0.3.1, 2026-05-23 Saturday (Calibration #23 substance refill: 44
→ ~100 lines + Euler equations explicit + 3-level pedagogy expanded)